far from each other, so once we reach the optimal number of clusters, the improvement with adding another centroid is almost null), but there is often a point that's like a watershed: on its left the slope of the curve is closer to (or larger than) -45°, on its right it's closer to 0°.

Three are, of course, a few details to keep into account to successfully implement this method: first, since k-means is a randomized method, it's important to run it several times per value of k; then you can pick the best value among all the runs (and also store the clustering produced, as we do in the notebook on the book's repo), or the average or median value for the metric, depending on what's your goal[26]. Moreover, you need to carefully choose the best metric for your problem: you might want to minimize the distance of points inside one cluster, to make sure clusters are as homogeneous as possible, or for instance maximize the distance between different clusters, to make sure to have a neat separation between different groups.

Listing 12.10 summarizes the steps that should be performed to successfully apply the elbow method (up to the plotting, not included for obvious reasons). As mentioned, you can check it out in this notebook on the book's repo.

Listing 12.10 The elbow method

```
function kmeansElbow(points, ksToTest, maxIter,
runsPerK)            #1
  results ←
{}                                                    #2
  for k in ksToTest do
         #3
```